Building Agent Hub with AI

From idea to Files
Communicating with AI, Defining standards, and Iterating

SPEAKER: **XINGYU YOU** DATE: **2026-10-20**

01 · FROM LAST TIME

From reliable AI work to Agent Hub

Continue the journey from our September share.

0918 · MAKE ONE TASK RELIABLE

01
Chat
Explore an idea

→

02
Structured Prompt
Define the task

→

03
Context and Evidence
Keep it reviewable

THIS TIME · DESIGN COORDINATED WORK

→

04
Workflow
Connect the steps

→

05
Agent
Bound each role

→

CASE
Agent Hub
Coordinate roles

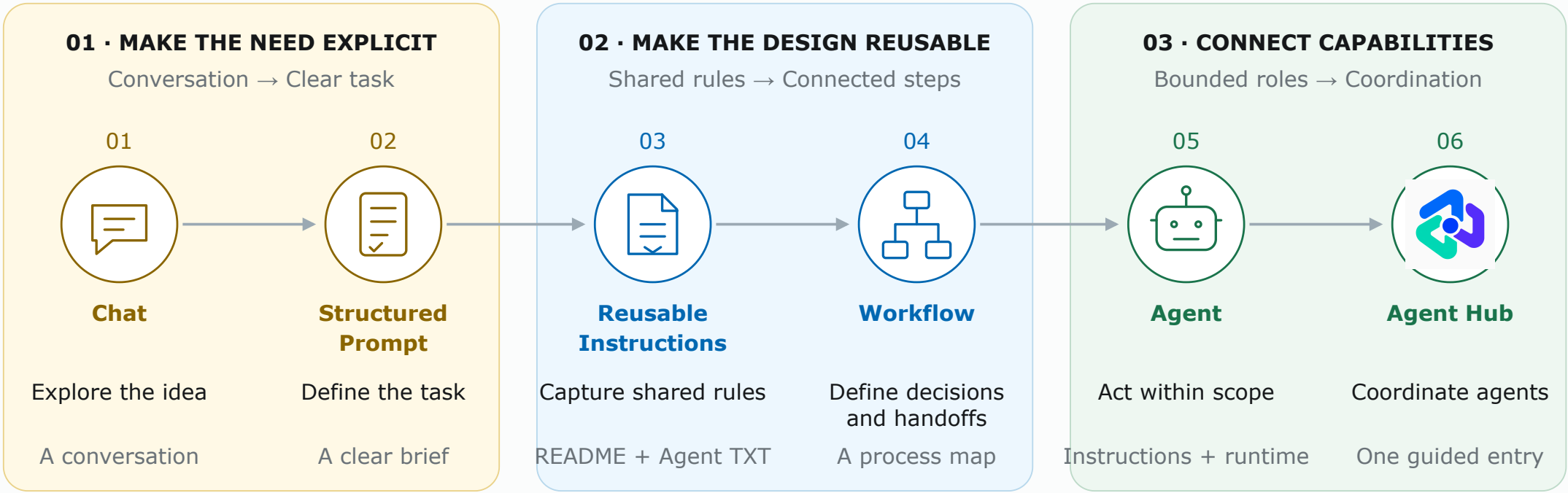
TODAY'S QUESTION

Who does what, what passes between roles, and where should the work stop?

01 · THE BIG PICTURE

From Chat to Agent Hub: the whole journey at a glance

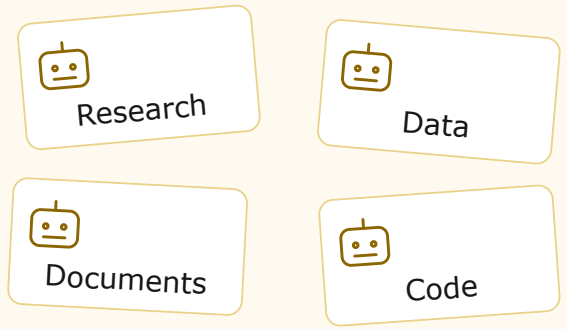
Follow one project from an initial conversation to reusable rules and coordinated agent roles.



A prompt becomes part of an agent design. Runtime, tools, state handling, and verification make execution possible.

Why Agent Hub? From too many options to coordinated work

01 · Too many agents



"Which one fits my task?"

The user knows the goal, not the agent.

02 · Recommend and explain

1 **Candidate A** Strong fit
Capability · input fit · permissions

2 **Candidate B** Partial fit
Capability · input fit · permissions

3 **Candidate C** Limited fit
Capability · input fit · permissions

Explain the ranking and limits.

Illustrative shortlist, not actual scores.

03 · Help complete the task

Select agents + plan

↓

User confirms scope

↓

Execute within permission

↓

Return reviewable results

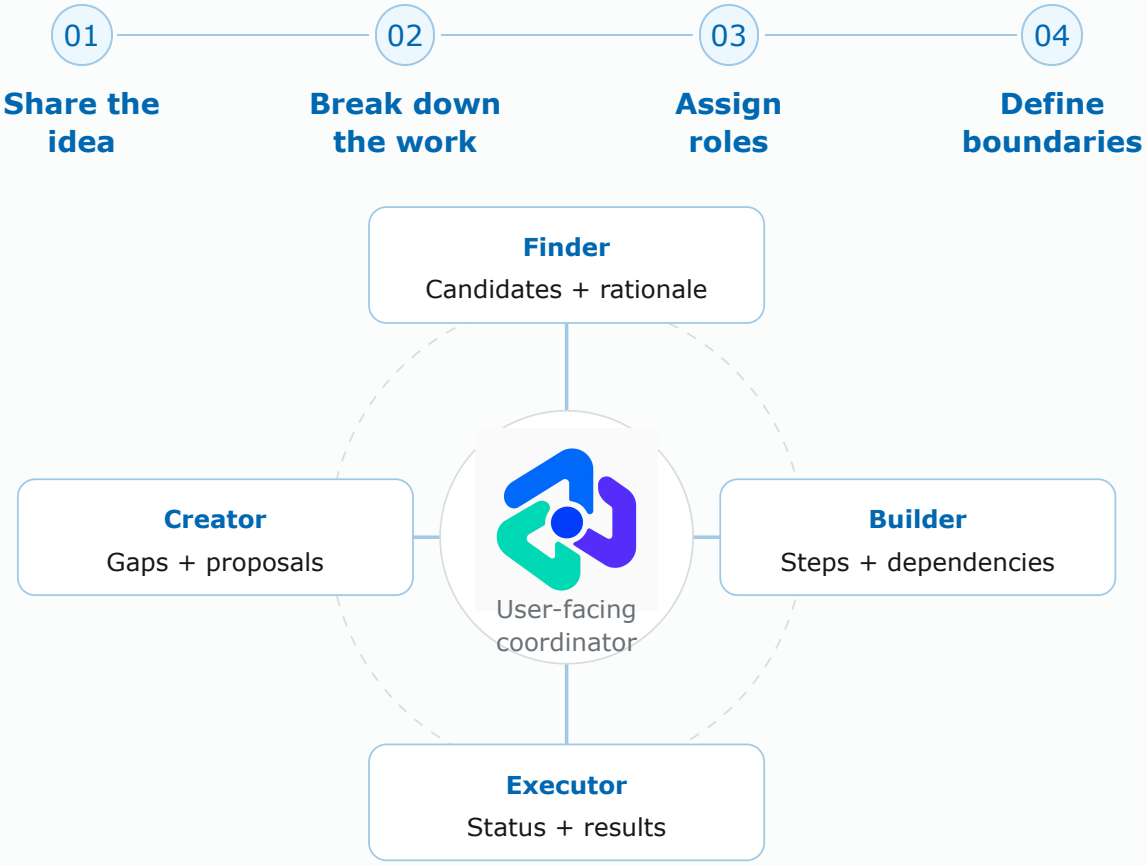
Hub coordinates the handoffs.

Agent concepts through a clinical-programming analogy

	Analogy to Clinical Programming Procedure
Agent	**一个标准 macro 加上它的 header 说明。** 写明用途、输入参数、输出和使用限制，只在说明范围内工作，超出范围的需求不会处理。
Registry	**Macro library 的 catalog。** 每个条目记录状态、功能、需要的输入和适用范围。列在 catalog 里只代表能被找到，不代表已经在你的 study 里验证过、可以直接运行。 状态：能用、测试中，还是已停用 功能：它能做什么，例如核对 spec 和 define.xml 需要的输入：运行前要提供什么，例如 spec 文件、define.xml 适用范围：能处理哪些情况，例如只支持 ADaM、不支持 SDTM
Agent Hub	**提交编程需求的统一入口，类似 programming lead 接需求。** 你只需要说明目标，例如"准备 DMC delivery：核对 spec 变更、对比新旧输出，并起草给 study team 的说明"。Hub 负责理解需求，找到合适的 Agent，安排先后顺序，再把推荐和计划带回来给你确认。
Finder	**在 macro library 里按需求挑合适的 macro。** 它会说明推荐理由和限制，并提醒你还需补充什么信息，例如"请提供要核对的 spec 版本"。只负责推荐，不会直接运行。
Builder	**写 programming plan，理清先后顺序和依赖关系，** 就像先 SDTM、再 ADaM、最后 TLF。只有任务需要多个步骤时才会用到，简单任务会直接跳过。
Blueprint	**这次任务的 run order 草案，类似 batch driver program。** 写清每一步由谁做、用什么输入、产出什么，以及依赖哪一步。确认之后才会执行。
Executor	**按批准的 run order 执行 batch job。** 只运行确认过的步骤，结果按 completed / failed / skipped 报告。失败后不会自动重跑整个流程。

02 · ROLE DESIGN

How we shaped the main-agent framework with AI



ILLUSTRATIVE CONVERSATION · NOT ORIGINAL CHAT

DESIGNER · Share the idea

I want a single entry point that recommends agents, explains fit, and helps complete tasks. Help me structure the system first.

AI · Break down the work

We need request understanding, capability matching, workflow planning, task execution, and capability-gap handling.

DESIGNER · Ask for the framework

Which work belongs to the main agent? Which needs a functional agent? Keep planning separate from execution and make boundaries explicit.

AI · Propose the role split

Hub owns interaction and routing. Finder matches, Builder plans, Executor runs confirmed work, and Creator assesses gaps and reuse.

03 · WORKFLOW

From role design to runtime handoffs: who does what?

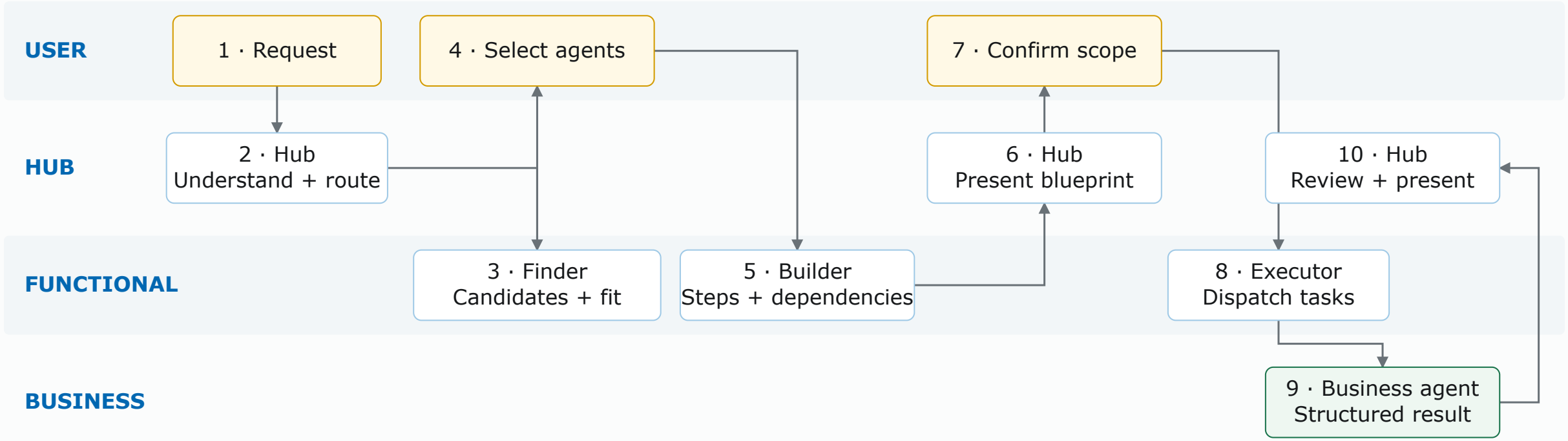
DESIGNER — AI · Make the responsibilities explicit

Designer and AI define this process; the USER lane represents the person using Agent Hub. For each step, define:

Owner: Who is responsible for this step?

Handoff: What does it receive, produce, and pass to the next role?

Boundary: When should it pause or return? Keep planning separate from execution.



From roles to an initial workflow



RECONSTRUCTED EXAMPLE · NOT ORIGINAL CHAT

DESIGNER · Connect the roles (D)

Using the agreed roles, outline the main path from a user request to a reviewable result.

AI · Propose a main path (AI)

Understand the request → find or validate agents → plan if needed → confirm → execute → review.

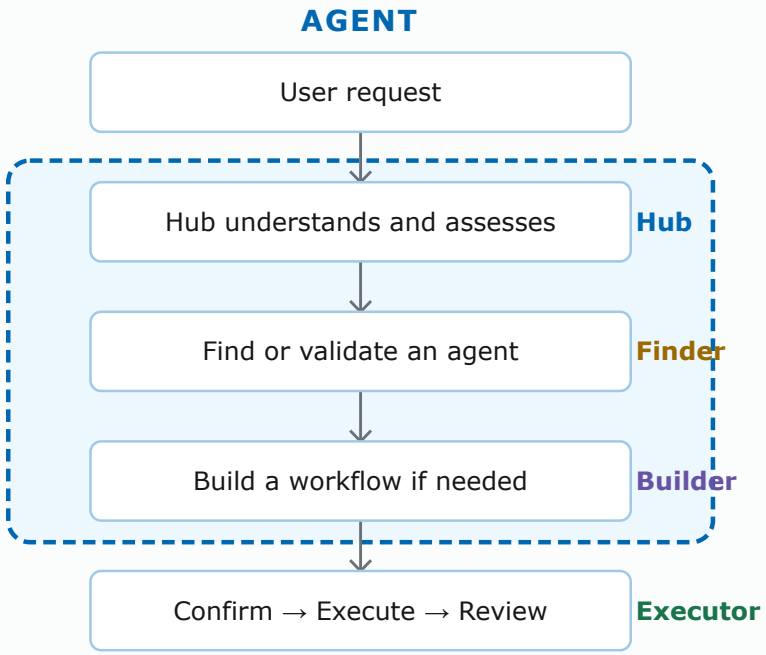
DESIGNER · Challenge the first draft (D)

Now examine it from the user's perspective. What different questions or missing inputs could change this path?

DESIGN OUTCOME

The main path is a starting point. Next, user questions reveal the missing branches.

From roles to an initial workflow



Programming lead
Receives the request, routes it to the appropriate roles, and brings recommendations or a plan back for user confirmation.

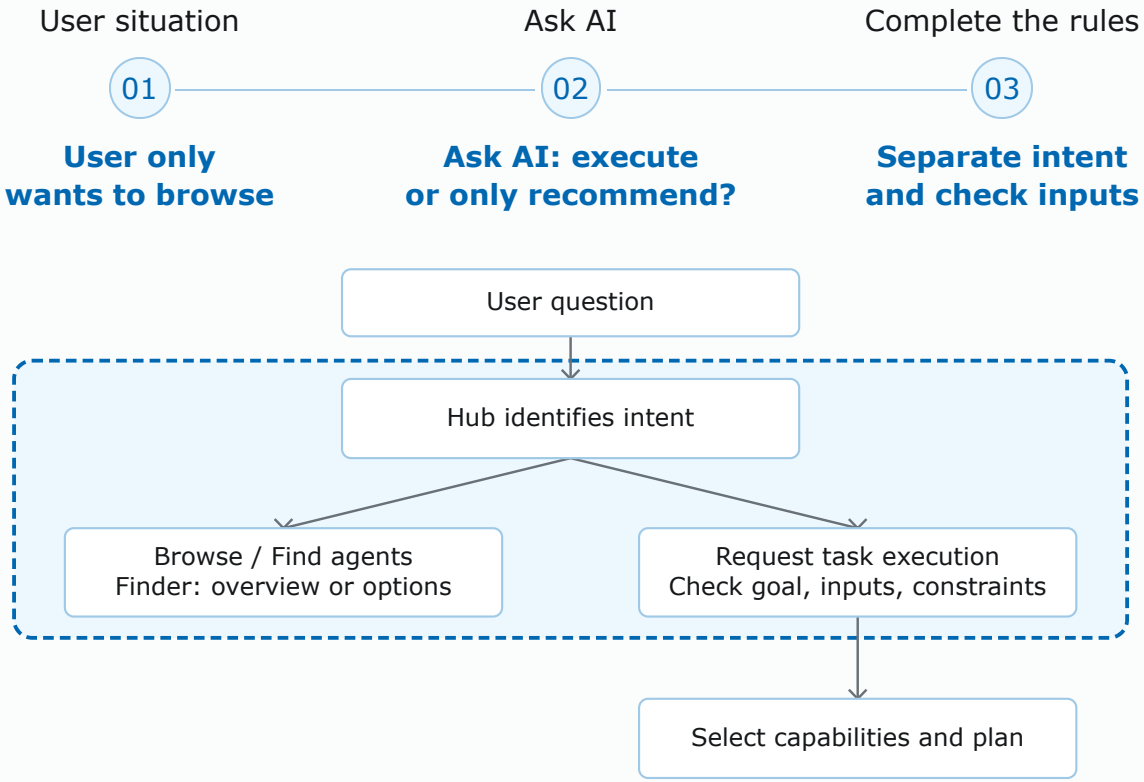
Macro library search
Searches the catalog, explains a macro's fit and limits, and identifies missing inputs; recommends but does not execute.

Programming planner
For multi-step work, sets step order, inputs, outputs, and dependencies (SDTM → ADaM → TLF); simple tasks skip planning.

Batch-job runner
Runs only confirmed steps and reports completed, failed, or skipped; failures do not trigger a full automatic rerun.

03 · WORKFLOW

User perspective 1: “I am only browsing.”



DESIGNER-LED FOLLOW-UP · ILLUSTRATIVE DIALOGUE

DESIGNER · Design question (D)

If a user only asks what agents exist, should Hub execute anything?

AI · Proposed rule (AI)

No. Separate browsing from execution. Finder returns an overview or recommendations.

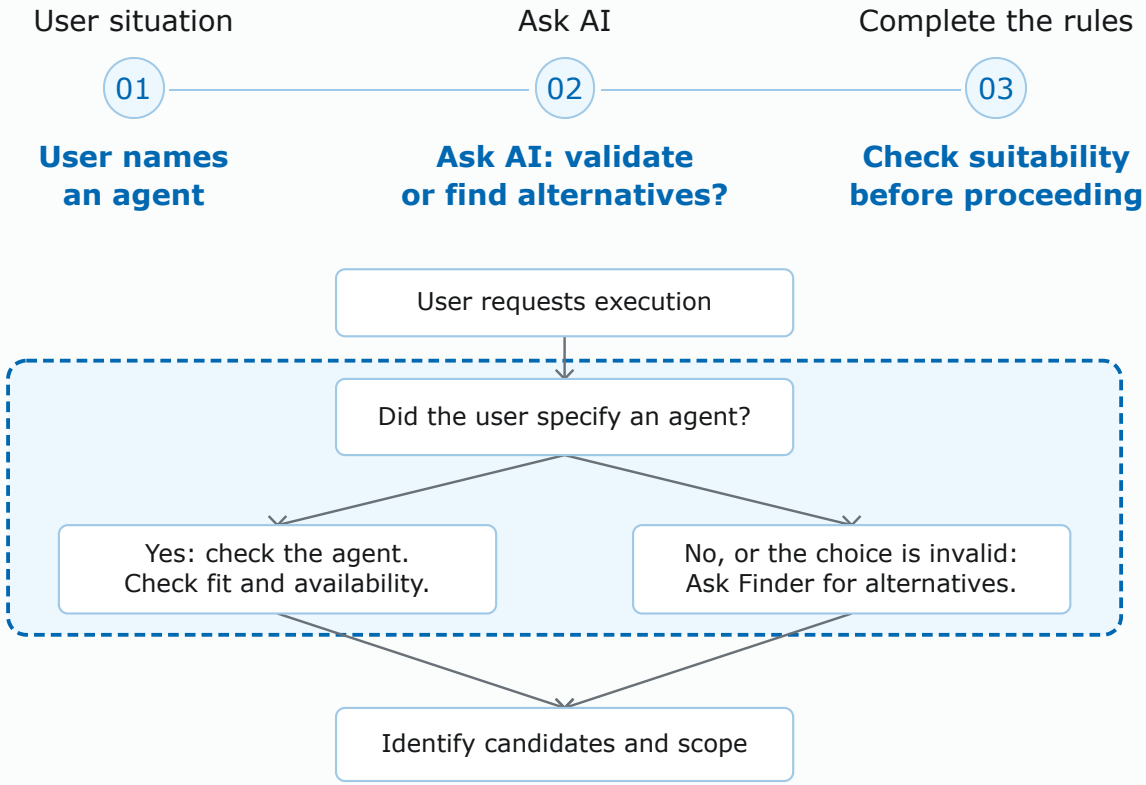
DESIGNER · Follow-up (D)

What if the goal is clear, but the user has not supplied the materials?

IMPLEMENTED IN THE DESIGN

Discuss suitable capabilities if useful, but wait for required inputs before execution. Ask focused questions if the goal itself is unclear.

User perspective 2: “I already chose an agent.”



DESIGNER-LED FOLLOW-UP · ILLUSTRATIVE DIALOGUE

DESIGNER · Start with the need (D)

Users may name an agent. Should Hub still ask Finder to recommend one?

AI · Proposed design (AI)

Validate the specified agent first. If it fits and can run, skip a fresh recommendation.

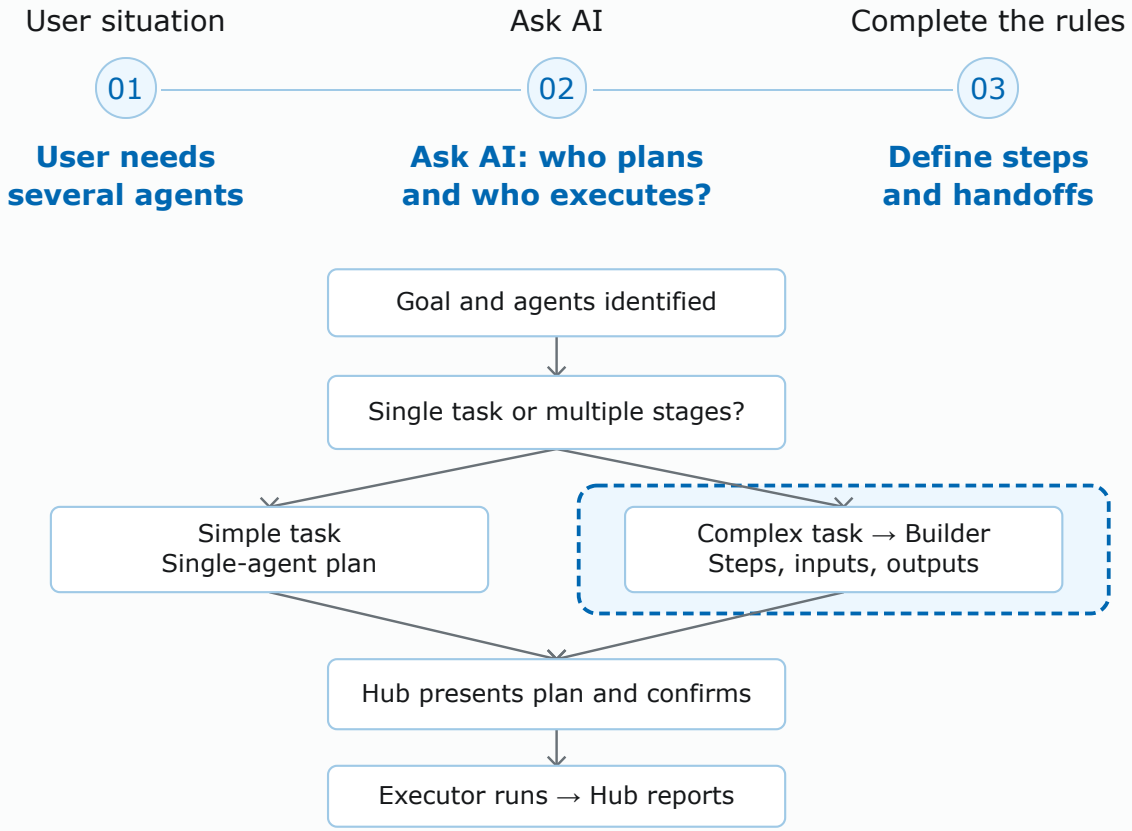
DESIGNER · Refine the boundary (D)

What if it is invalid? Does choosing an agent authorize execution?

IMPLEMENTED IN THE DESIGN

Invalid choice → explain and ask Finder. Selection alone does not authorize execution. Scope confirmation remains required.

User perspective 3: “My task needs several agents.”



DESIGNER-LED FOLLOW-UP · ILLUSTRATIVE DIALOGUE

DESIGNER · Design question (D)

How should several agents work together on one complex task?

AI · Proposed rule (AI)

Break the goal into ordered steps. Specify each step’s owner, inputs, outputs, and dependencies.

DESIGNER · Follow-up (D)

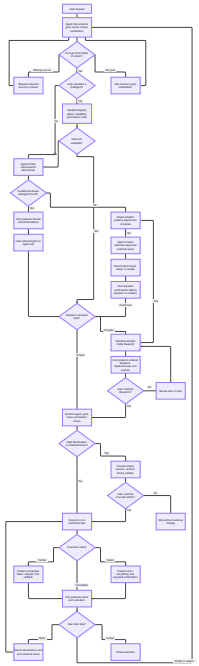
Who plans, who executes, and what exactly passes between them?

IMPLEMENTED IN THE DESIGN

Builder creates the blueprint. Hub presents it for confirmation. Executor receives the confirmed plan and dispatches work within scope.

03 · COMPLETE WORKFLOW

Ask AI to connect the main path, branches, and return paths



Click to explore the full diagram · zoom + / −

DESIGNING THE PATH · ILLUSTRATIVE DIALOGUE

DESIGNER · Connect the pieces D

Combine our main path and user-question branches into one end-to-end flow. Show where each branch rejoins, waits, or stops.

AI · Trace the return paths AI

I'll connect the normal path, clarification loops, plan revisions, and result review. I'll identify missing connections rather than invent a decision.

DESIGNER · Challenge the map D

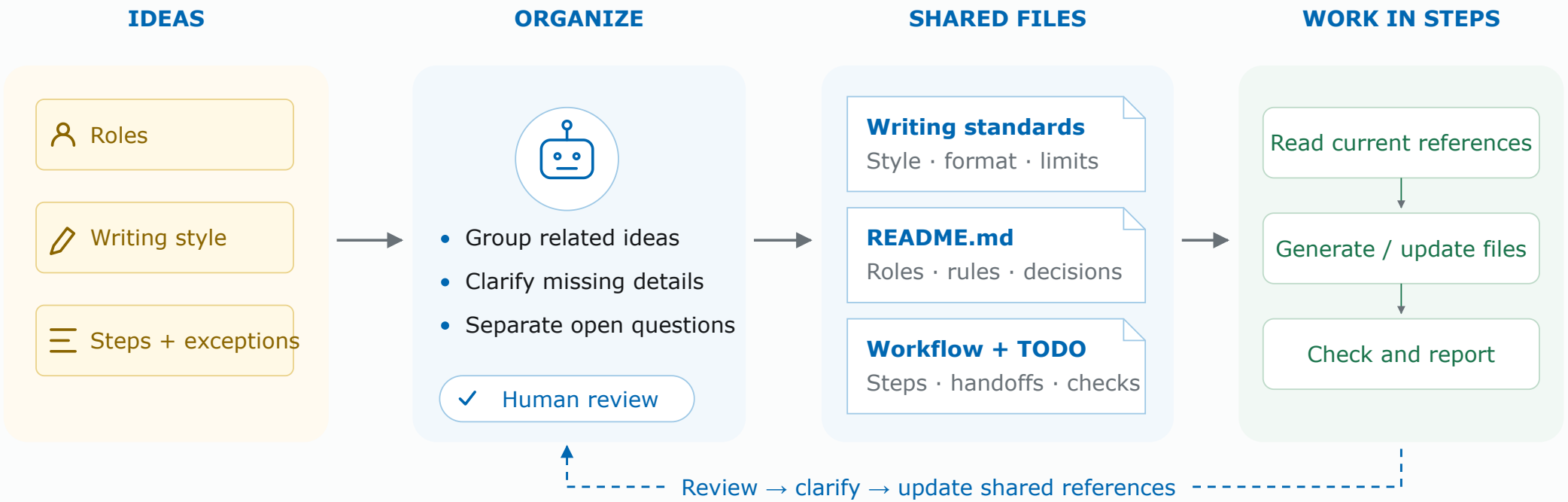
If inputs are missing, the plan is rejected, or execution fails, where do we go next? Check each branch for a clear return or stopping point.

Building a flowchart like this helps us check with AI that every branch has a clear next step, return path, or stopping point.

Actual project diagram referenced by README → End-to-End Agent Hub Flow. Complete project diagram shown for orientation. Open the zoomable view for readable labels and all branches. Copy canvas expanded to include source nodes beyond its original bounds.

04 · IMPLEMENT WITH AI

Use AI twice: organize our ideas, then work from the agreed files



First: Ask AI to organize scattered ideas into clear roles, workflows, and shared standards.

Second: Ask AI to generate and refine content step by step using the agreed files.

04 · IMPLEMENT WITH AI

Four reusable files, four clear jobs

We save the agreed design in separate files, each with a clear purpose: **system design**, **definition standards**, **capability discovery**, or **next steps**. Tell AI which reference to read and which file to update so it can build on our agreement across tasks.

README.md

Shared Design

Roles · Rules · Decisions

Business README

Definition Pattern

Name · Description · Instructions

Registry

Find Capabilities

Tags · Fit · Availability

TODO.md

Track Next Steps

Tasks · Checks · Open Items

Registry · Agent-Registry.txt

Status
Active · Testing · Deprecated

Capability
e.g., spec and define.xml checks

Required inputs
spec · define.xml

Scope
ADaM, not SDTM

Listed means discoverable; it does not mean study-validated or ready to run.

04 · IMPLEMENT WITH AI

Ask AI to summarize our agreement, then inspect the README

01 · Ask AI to summarize

DESIGNER · Summarize and save

Summarize our agreed roles, workflow, and rules in README.md. Preserve our decisions and flag unresolved questions.

AI · Organize the agreement

I'll group the content into the shared sections, record accepted choices in Decision Log, and keep unresolved items in Open Questions.

02 · Use a fixed framework

Responsibilities

Core Architecture
- ↳ Agent Hub
- ↳ Finder / Builder
- ↳ Executor / Creator

Workflow Rules

Task Routing Logic
- ↳ Browse / Recommend
- ↳ Simple / Complex

Decision Record

Decision Log
- ↳ Accepted Choices

Open Questions
- ↳ Unresolved Choices

03 · Inspect the actual file

agent hub / README.md

Architecture | Finder | Standard | Decisions

Agent Finder

Agent Finder discovers and recommends suitable agents from the enterprise-certified and recommended Subagent resource library.

Use Agent Finder when:

- The user wants to know what agents exist.
- The user searches for agents by category, keyword, goal, or business scenario.
- The user wants the best agent for a task.
- The user wants to execute a task but has not specified the target agent.

Use the business-agent README as a generation standard

business-subagents/README.md

Definition Format

Name + Description
- ↳ English / Chinese Names
- ↳ Description ≤ 800 Characters

Instruction
- ↳ Roles / Triggers / Inputs
- ↳ Outputs / Rules / Limits

Shared Boundaries

Structured Results
- ↳ Return Upstream Only
- ↳ Missing Inputs: needs_clarification
- ↳ assumptions / missing_fields

Consistency Checks

Registry + TXT
- ↳ Keep Metadata Aligned

EXAMPLE · Presentation-Rehearsal-Coach-Agent.txt

Presentation Rehearsal Coach Agent

英文名称：Presentation Rehearsal Coach Agent

中文名称：演讲排练反馈助手

Description

Reviews supplied presentation attempts for clarity, structure, audience fit, and estimated timing, with explicitly simulated audience questions. Use when improving clarity, structure, audience fit, and estimated timing through simulated questions. Returns the requested communication draft or one practice turn with clear assumptions and review notes.

Instruction

Rules summarized from business-subagents/README.md. Right: the actual project definition, available to scroll.

05 · BUILD AND ITERATE

Build one agreed step at a time, then update TODO

DESIGNER · Control the implementation pace

Read the current README and TODO. Propose the implementation sequence first. This round, only refine the Hub and Finder definitions. Report checks and pending decisions before continuing.

01



Read the references

README + TODO

Roles · format · open tasks

→

02



Set this round's scope

Hub + Finder only

Name the files and outputs

→

03



Generate and review

Draft → check → revise

Compare with agreed rules

→

04



Save the next starting point

Update TODO

Done · open · next action

After each round, report the files changed, checks performed, and open questions. If the next step requires changing agreed responsibilities or rules, or expanding scope, raise the issue before proceeding.

05 · BUILD AND ITERATE

Resume without losing context or redesigning agreed work

Why restore context?

Lost Context
Earlier decisions are missing from the next request.

Repeated Design
Agreed roles are proposed all over again.

Change Spreads
A small revision rewrites unrelated branches.

01 Read References — 02 Restore Agreement — 03 Locate the Change — 04 Continue Within Scope

READ FIRST README | Workflow | Agent TXT | TODO

DESIGNER · Restore context (D)

Read the current README, relevant workflow, Agent TXT, and TODO. Summarize the agreed roles, key rules, completed work, and open items before continuing.

AI · Locate the change (AI)

I'll locate the branch affected by this request, identify the files involved, and explain the proposed change. If the references conflict or a decision is missing, I'll flag it first.

DESIGNER · Limit the revision (D)

Update only the agreed scope and preserve unrelated branches. Report the files changed, checks performed, and remaining questions. Ask before changing an accepted role or rule.

05 · BUILD AND ITERATE

Check a revision by walking one concrete request

01 **Choose a request**

CHALLENGE REQUEST

“Use this agent. I’ll send the files later.”

The agent is selected, but required inputs are not ready.

02 **Trace the branch**

Validate the agent
Suitable for this goal?

↓

Check required inputs
Missing materials → pause

↓

Return the missing fields
Hub obtains the needed input

03 **Compare the files**

Compare the written rules

README + Hub
Does routing preserve the input check?

Relevant agent TXT
Are missing fields and pause conditions explicit?

04 **Fix the mismatch**

Targeted feedback to AI

If the files disagree, identify the conflicting clauses and propose a focused correction.

Revise the affected files, then walk through the same request again.

05 · ITERATE & VALIDATE

After updating the files, verify what is actually complete

Updating all the files does not mean the feature is ready. Distinguish a completed definition from verified execution and results.

01 · Definition

- Roles and rules are consistent
- TXT and Registry agree

Written definition checked

→

02 · Runtime

- Tools and permissions configured
- Actual agent calls observed

Run evidence needed

→

03 · Results

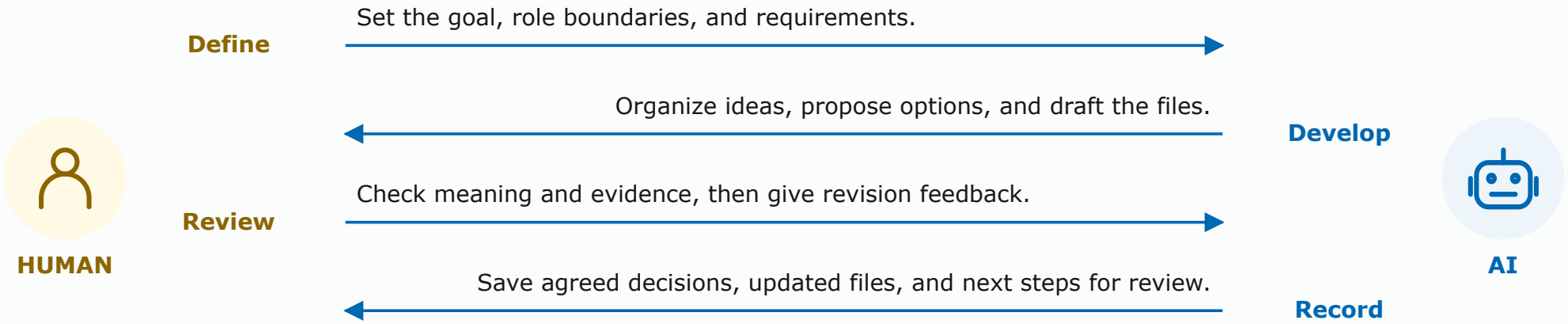
- Outputs supported by evidence
- Acceptance criteria reviewed

Output review needed

Actual project correction · TODO, 2026-10-15

`Active` means the TXT definition is complete and discoverable. It does not establish deployment or runtime validation.

Human judgment and AI assistance at each iteration



Human judgment guides each iteration.

AI helps organize, develop, and record the work.

06 · PRACTICAL TAKEAWAYS

A reusable communication template: design the workflow and README

01 Goal
What should this achieve?

02 Context
Which ideas and files should AI use?

03 Scope
What should AI produce this round?

04 Constraints
Which rules and boundaries must remain?

05 Acceptance
What outputs and checks define completion?

REUSABLE EXAMPLE · ONE COMPLETE REQUEST TO AI

Goal. Help me design Agent Hub, a single entry point for finding and coordinating business agents.

Context. Use my requirements and available project files. First summarize accepted decisions and open questions.

Scope. Map likely user requests into a workflow. Define Hub, functional, and business-agent responsibilities. Propose a minimal file structure and draft the README: vision, architecture, routing, definition standards, decisions, and open questions.

Constraints. Keep recommendation, planning, and execution separate. Require confirmation before execution. Flag missing decisions. Do not invent tools, deployment, or test results.

Acceptance. Return the workflow diagram, file responsibilities, and README draft. Walk through representative user questions and list conflicts, assumptions, and next steps.

01 · Fill the five elements
Start with one goal and the current references.

02 · Bound this round
Ask for a workflow and README draft first.

03 · Walk one request
Check owners, missing inputs, handoffs, and stops.

TAKEAWAY

From a conversation to a shared, reviewable design

01

Design the logic

Define roles, decisions, and handoffs before expanding the system.

02

Keep a shared reference

Capture accepted rules in README, diagrams, and agent instructions.

03

Review every change

Check affected files and separate written definitions from verified execution.

Capture the idea as context.

Capture requirements as standards.

Use files to guide AI through implementation and review.